

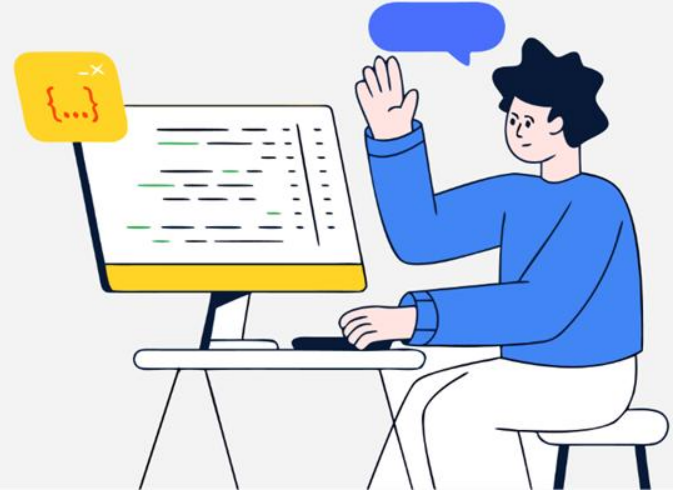
Guidelines

- Kindly use the given template for submitting your prototype(Make a copy of the template)
- One team is only required to submit one project.
- Follow this template while preparing your presentation.
- You are welcome to add as many POCs and design concepts to support your project.
- The project should be feasible and the team members should be capable enough, to come up with the fully functioning prototype of the same idea, if required.
- In case of queries, kindly reach out to us at solutionchallengesupport@hack2skill.com

Note: Prototype must include cloud deployment and use at least one Google AI model or service (e.g., Gemini) as part of the solution.



Solution Challenge



Team Details

- a. Team name: **Awesome_AI**
- b. Team leader name: **Yasmeen Begum**
- c. Problem Statement: **Rapid Crisis Response Accelerated Emergency Response and Crisis Coordination in Hospitality**

Brief about your solution

Hospitality Emergency Response Coordination (HERC) is a structured system for managing emergencies in hotels and resorts, where guest safety and brand reputation must be balanced. Unlike offices, hospitality environments face challenges such as transient guests unfamiliar with layouts and 24/7 operations. CrisisSync addresses these challenges through five pillars: instant detection and alerting via a centralized SOS portal, situational awareness with interactive floor maps, resource management through personnel tracking, AI-driven decision support that reduces cognitive load, and synchronized communication across multiple platforms to avoid fragmented messaging. Traditional responses rely on walkie-talkies and binders, but CrisisSync digitizes the Emergency Response Plan into a dynamic ecosystem. Before incidents, AI generates training drills; during crises, it enables real-time triage, deployment, and evacuation guidance; and after incidents, it automates reporting and analysis. This integrated approach ensures faster, smarter, and more coordinated emergency management in hospitality settings.

Opportunities

1.How different is it from any of the other existing ideas?

- **Lifecycle Coverage:** CrisisSync manages the full crisis journey — from SOS trigger to AI-generated post-incident reports — unlike passive panic buttons or siloed walkie-talkies.
- **Guest Integration:** It includes guests in the loop with real-time updates, while most systems only focus on staff communication.

2.How will it be able to solve the problem?

- **Centralized Hub:** The EOC ensures all stakeholders see the same data instantly, eliminating delays and miscommunication.
- **AI + Automation:** Gemma 4 provides deterministic action plans, and the n8n bridge broadcasts alerts across multiple channels in seconds.

3. USP of the proposed solution

- **Unified Crisis Intelligence Engine:** Combines personnel tracking, AI triage, and multi-channel automation in one hospitality-specific interface.
- **Visual + Real-Time Awareness:** Interactive maps with live overlays of incidents and staff ensure clarity and speed under pressure.

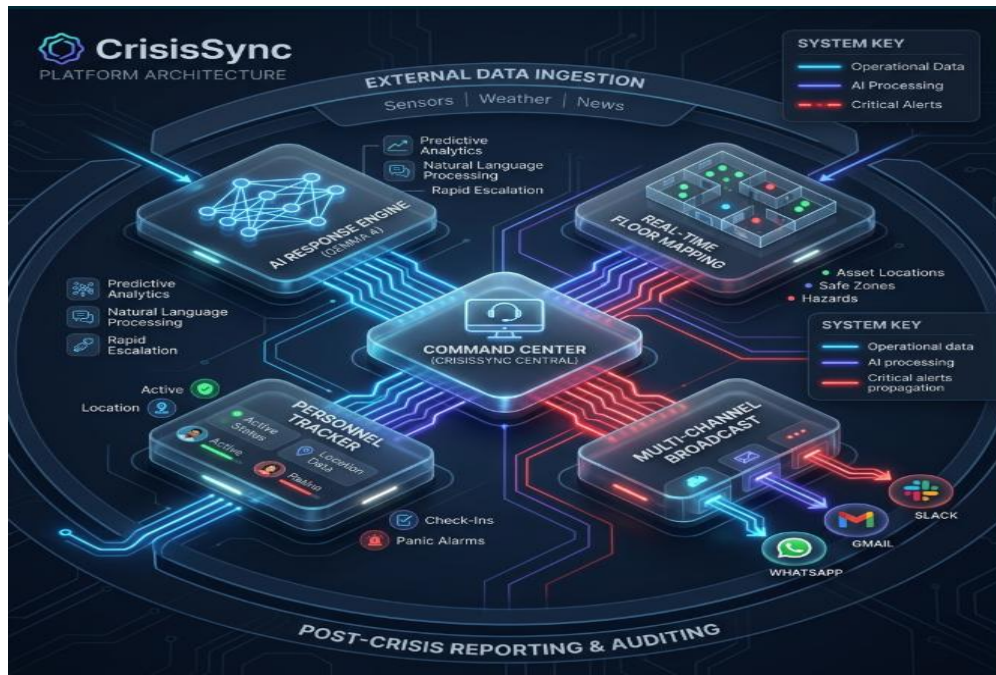
List of features offered by the solution

- 1.AI Crisis Assistant** – Gemma 4 analyzes live incidents and instantly provides response plans, evacuation steps, and communication templates.
- 2.Interactive Mapping** – Multi-floor maps show pulsing incident markers and real-time staff locations for precise coordination.
- 3.Multi-Channel Broadcasting** – The n8n bridge pushes alerts to WhatsApp, Gmail, and Slack at once, keeping everyone informed.
- 4.Instant SOS Portal** – Guests and staff can trigger immediate venue-wide emergency protocols without phone delays.
- 5.Personnel Tracking** – Monitors availability and location of responders, ensuring the closest qualified team reaches the scene fast.

Process flow diagram or Use-case diagram



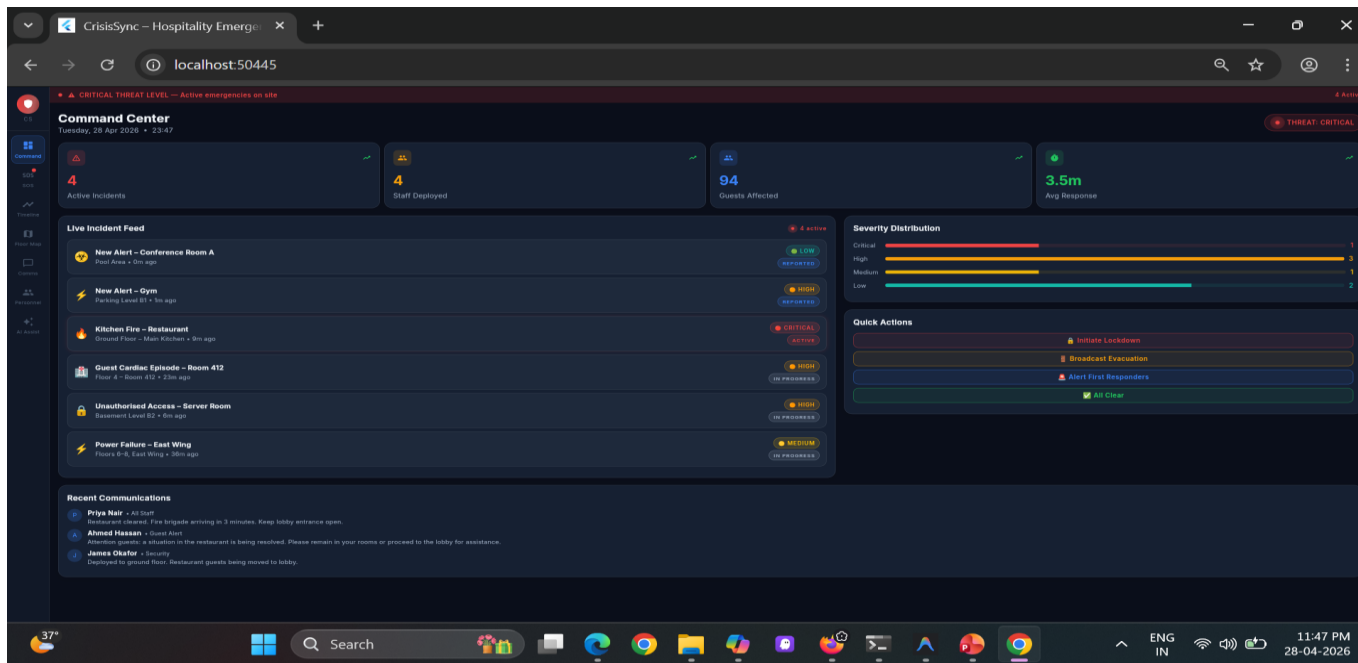
Architecture diagram of the proposed solution



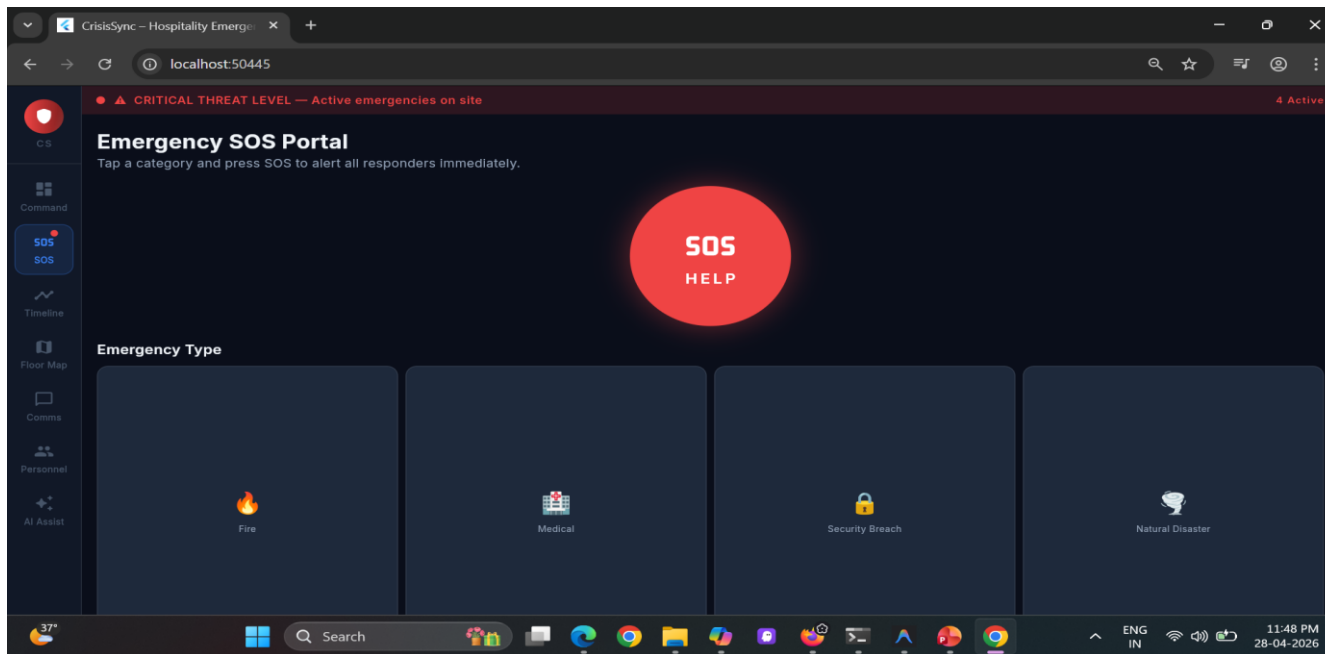
Technologies to be used in the solution

- Frontend:** Flutter (Windows, Web, Android, iOS)
- State Management:** Provider
- Backend:** Firebase (Firestore, Auth, Cloud Messaging)
- AI Engine:** Gemma 4 (HuggingFace Inference API)
- Integration Layer:** n8n (Automation Workflow)
- Protocol:** Model Context Protocol (MCP) for AI tool access

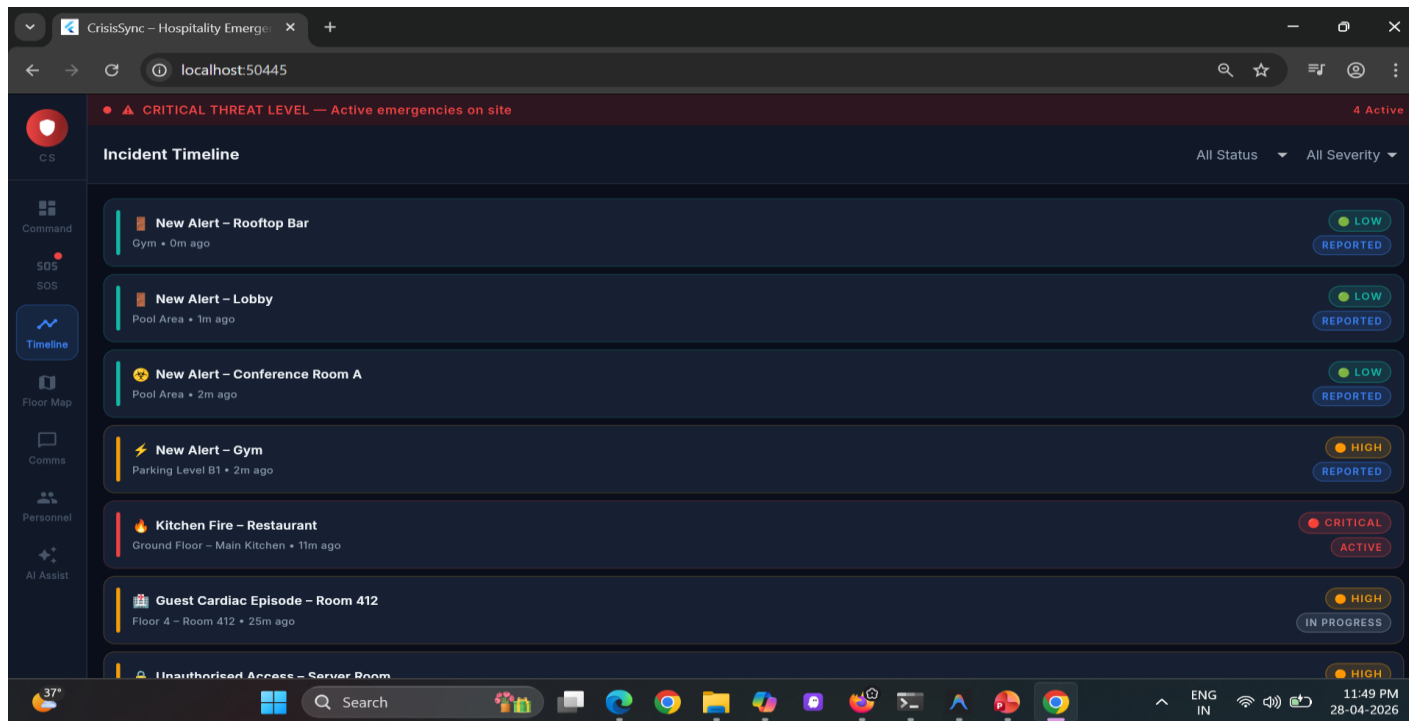
Snapshots of the MVP



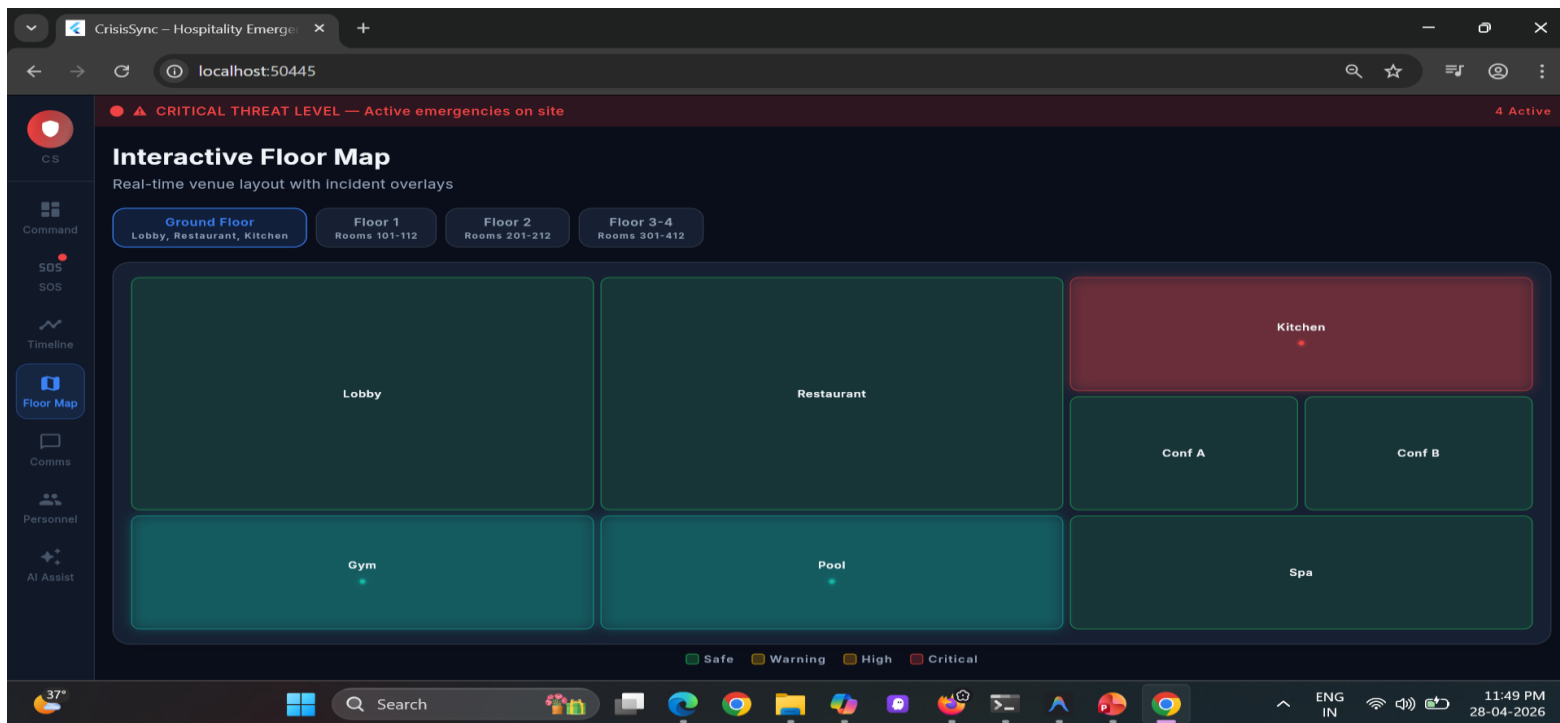
Snapshots of the MVP



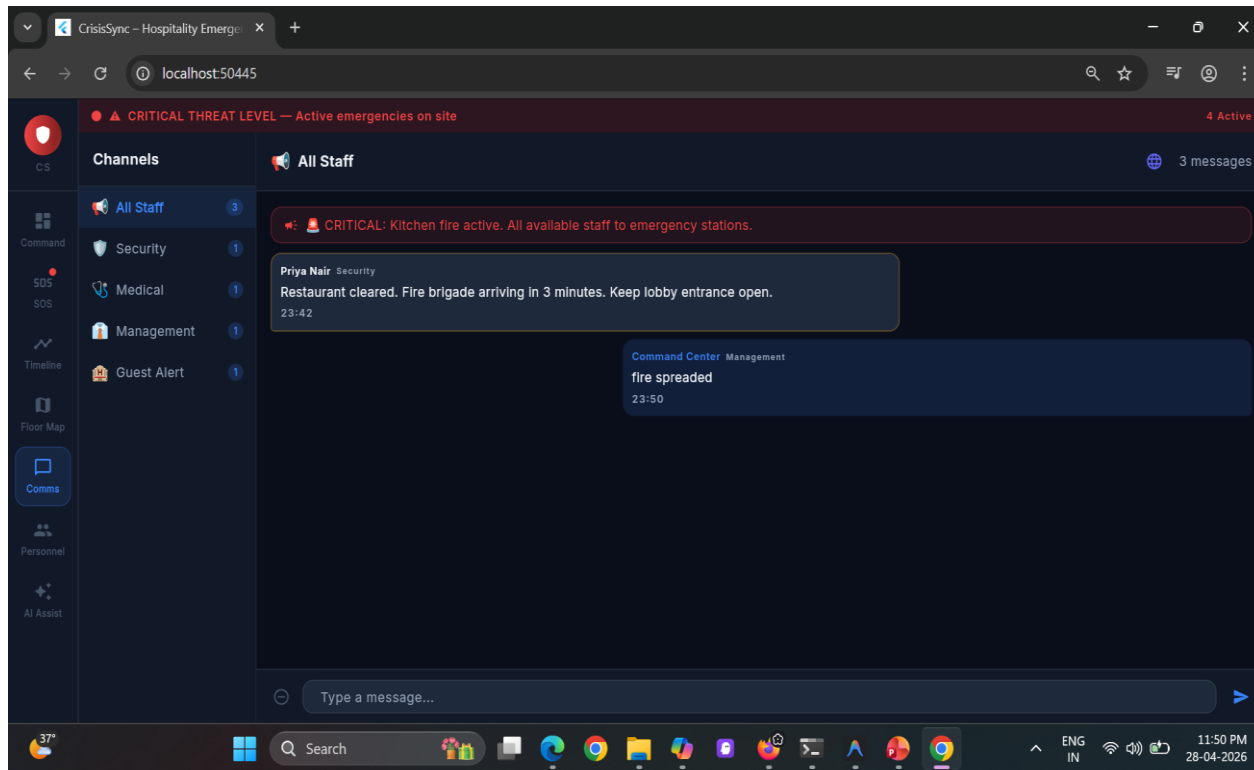
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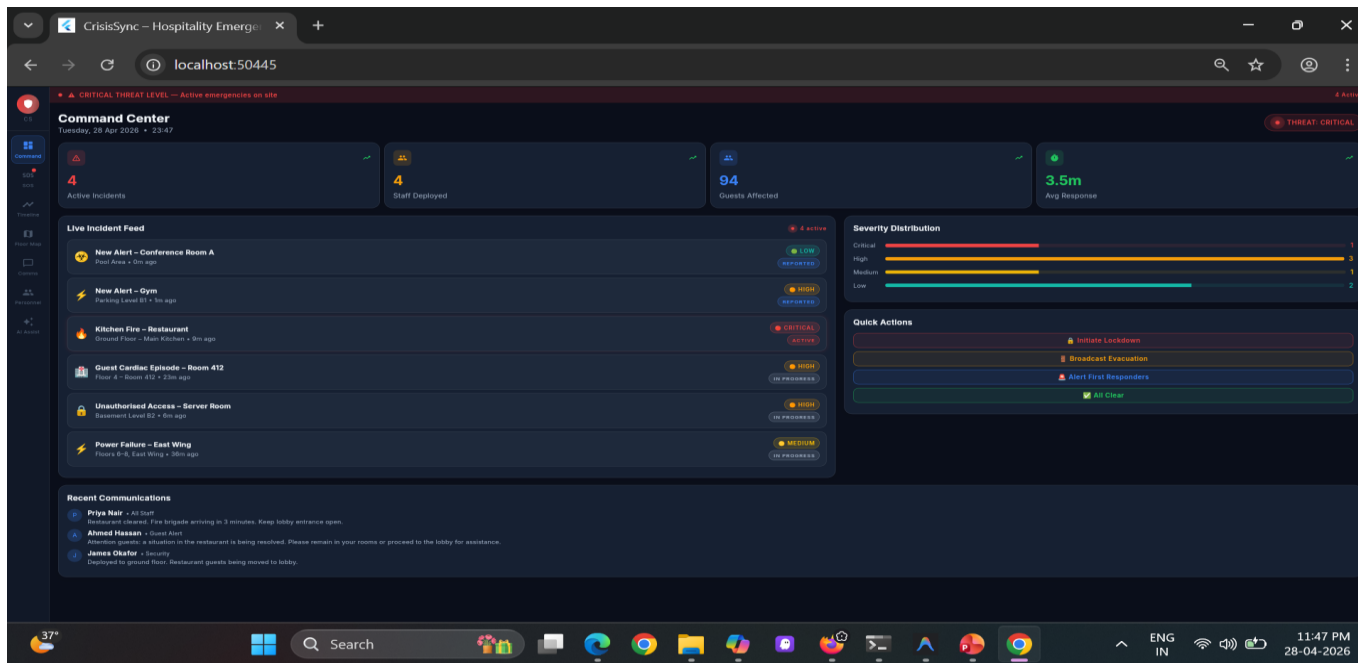
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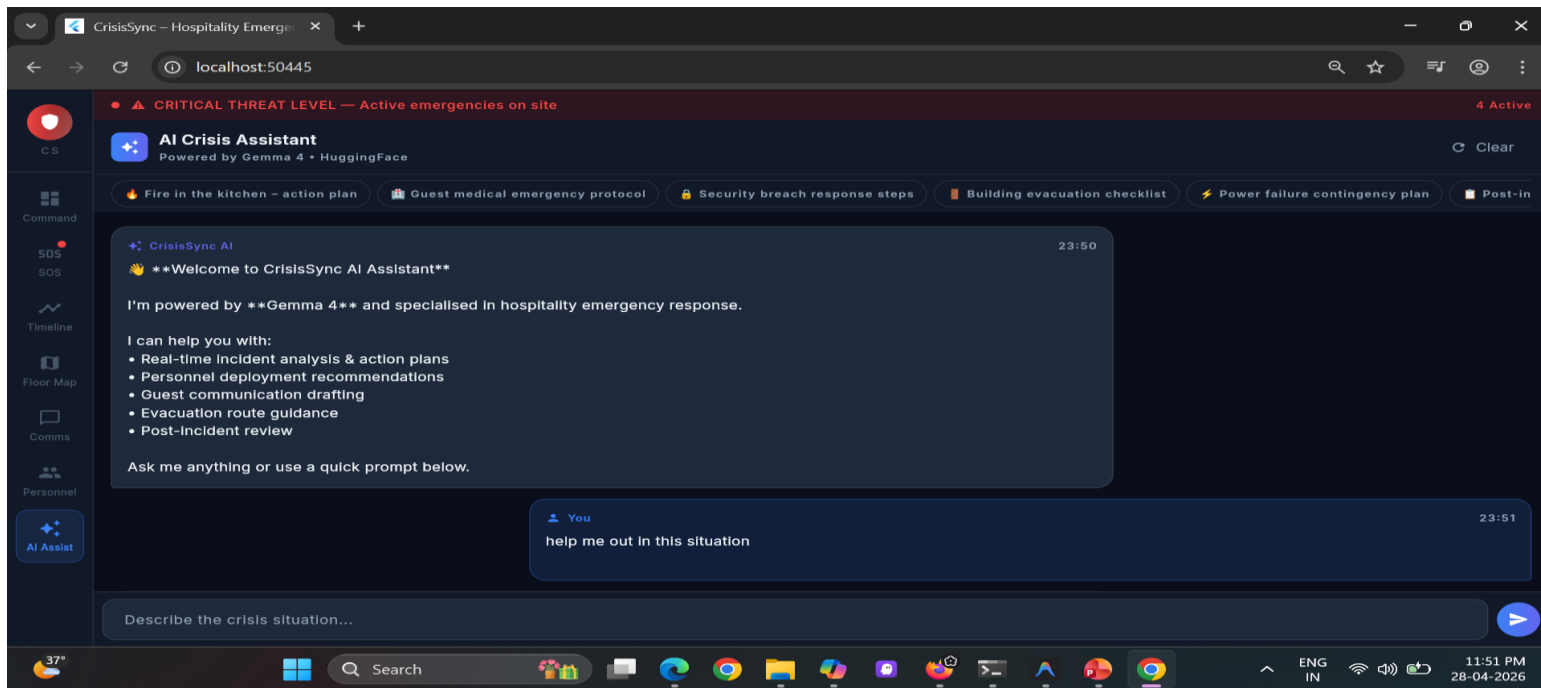
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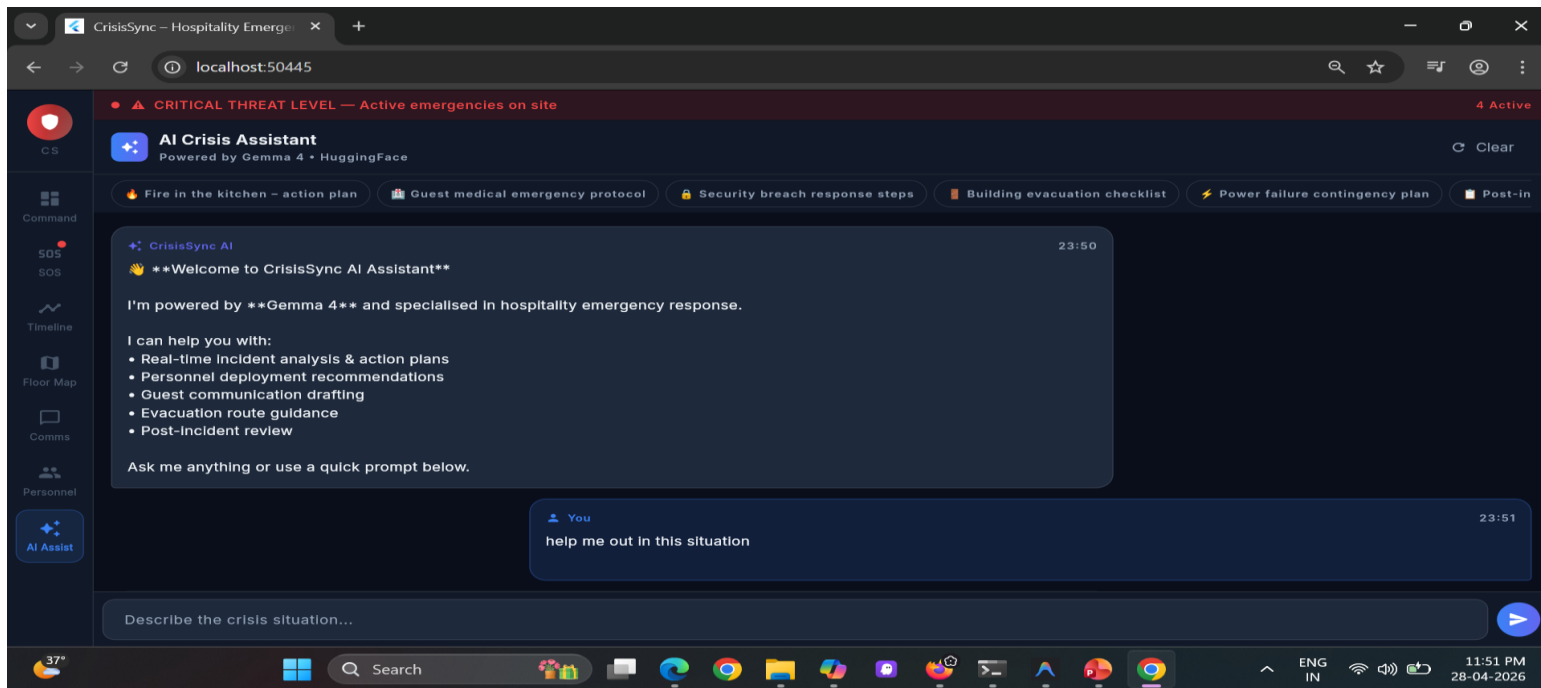
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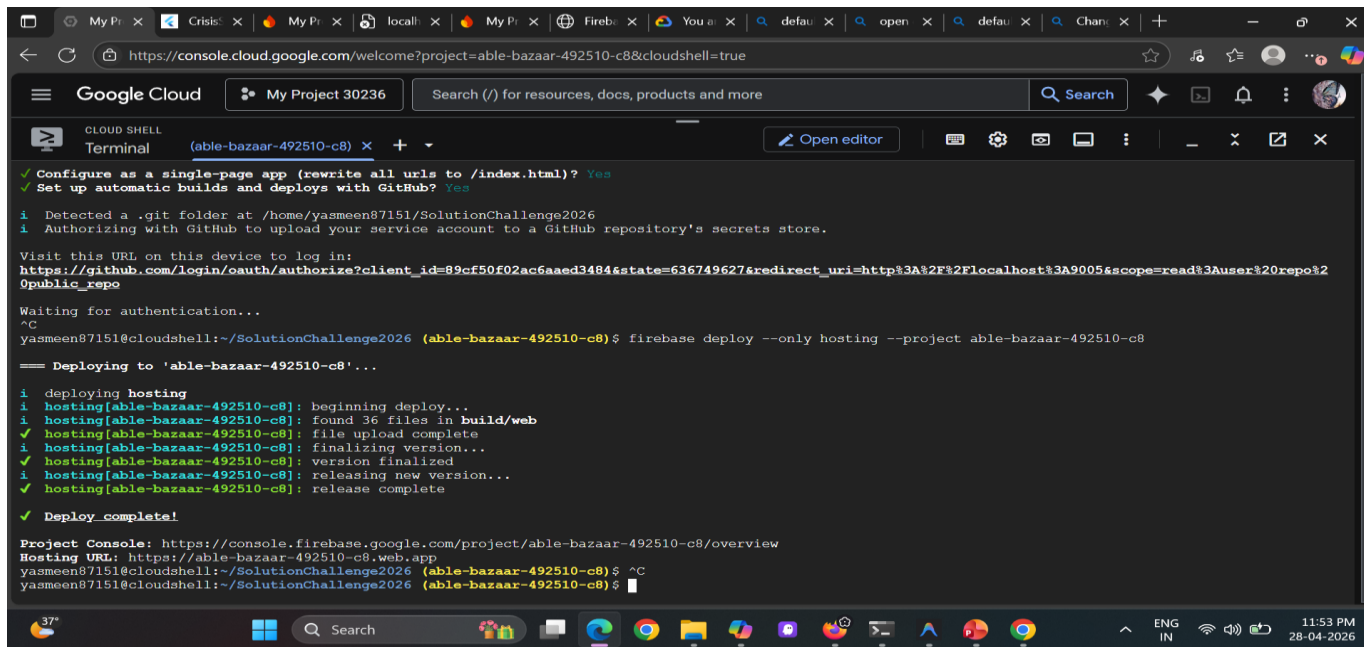
Snapshots of the MVP



Snapshots of the MVP



Snapshots of the MVP



The screenshot shows a Google Cloud console window with a terminal open. The terminal displays the following output:

```
✓ Configure as a single-page app (rewrite all urls to /index.html)? Yes
✓ Set up automatic builds and deploys with GitHub? Yes

i Detected a .git folder at /home/yasmeen87151/SolutionChallenge2026
i Authorizing with GitHub to upload your service account to a GitHub repository's secrets store.

Visit this URL on this device to log in:
https://github.com/login/oauth/authorize?client_id=89cf50f02acfaaed3484&state=636749627&redirect_uri=http%3A%2F%2Flocalhost%3A9005&scope=read%3Auser%20repo%20public_repo

Waiting for authentication...
^C
yasmeen87151@cloudshell:~/SolutionChallenge2026 (able-bazaar-492510-c8) $ firebase deploy --only hosting --project able-bazaar-492510-c8

== Deploying to 'able-bazaar-492510-c8'...

i deploying hosting
i hosting[able-bazaar-492510-c8]: beginning deploy...
i hosting[able-bazaar-492510-c8]: found 36 files in build/web
✓ hosting[able-bazaar-492510-c8]: file upload complete
i hosting[able-bazaar-492510-c8]: finalizing version...
✓ hosting[able-bazaar-492510-c8]: version finalized
i hosting[able-bazaar-492510-c8]: releasing new version...
✓ hosting[able-bazaar-492510-c8]: release complete

✓ Deploy complete!

Project Console: https://console.firebase.google.com/project/able-bazaar-492510-c8/overview
Hosting URL: https://able-bazaar-492510-c8.web.app
yasmeen87151@cloudshell:~/SolutionChallenge2026 (able-bazaar-492510-c8) $ ^C
yasmeen87151@cloudshell:~/SolutionChallenge2026 (able-bazaar-492510-c8) $
```

The terminal window is titled "Cloud Shell" and "Terminal". The browser address bar shows the URL: <https://console.cloud.google.com/welcome?project=able-bazaar-492510-c8&cloudshell=true>. The Google Cloud logo and "My Project 30236" are visible at the top of the terminal interface.

Additional Details/Future Development (if any)

- 1. IoT Integration** – Smart sensors and automated locks for faster detection and venue lockdowns.
- 2. Offline Resilience** – Mesh networking and local edge computing to keep coordination alive during outages.
- 3. Advanced AI** – Real-time translation plus sentiment/audio analysis to interpret guest alerts more accurately.
- 4. Wearables & Biometrics** – Health monitoring for responders and NFC guest badges for one-tap SOS with location tracking.
- 5. Training & Digital Twin** – Simulation mode for staff drills and 3D property visualization for superior situational awareness.

Provide links to your:

1. GitHub Public Repository- <https://github.com/Yasmeen-Begum/SolutionChallenge2026>
2. Demo Video Link (3 Minutes)-
3. MVP Link-<https://console.firebase.google.com/project/able-bazaar-492510-c8/overview>
4. Working Prototype Link-[CrisisSync – Hospitality Emergency Response](#)



Solution Challenge



Thank You